The due date for submitting this assignment has passed.
Due on 2026-03-11, 23:59 IST.

You may submit any number of times before the due date. The final submission will be considered for grading.

You have last submitted on: 2026-03-11, 08:07 IST

Note: This assignment will be evaluated after the deadline passes. You will get your score 48 hrs after the deadline. Until then the score will be shown as Zero.

Multiple Select Questions (MSQ)

1) Which of the following statements are correct? ***1 point***

- [ ] If A and B are two square matrices, then Rank(AB)=Rank(BA).
- [x] If A and B are two 2×2 square matrices, then Rank$(AB) \leq$ Rank(B).
- [x] If A and B are two 2×2 square matrices, then Rank$(AB) \leq$ Rank(A).
- [x] If A and B are $n \times n$ matrices of rank n, then AB has rank n.

Yes, the answer is correct.
Score: 1

Accepted Answers:

If A and B are two 2×2 square matrices, then Rank$(AB) \leq$ Rank(B).
If A and B are two 2×2 square matrices, then Rank$(AB) \leq$ Rank(A).
If A and B are $n \times n$ matrices of rank n, then AB has rank n.

2) Match the vector spaces (with the usual scalar multiplication and vector addition as in $M_{3\times3}(\mathrm{R})$) in column A with their bases in column B and the dimensions of the vector spaces in column C in Table : M2W4GA1. ***1 point***

	Vector space (Column A)		Basis (Column B)		Dimension of the vector space (Column C)
a)	$V = \left\{ \begin{bmatrix} x & y & z \\ 0 & z & x \\ y & 0 & 0 \end{bmatrix} \mid x + 2y = z, \text{ and } x, y, z \in \mathbb{R} \right\}$	i)	$\left\{ \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix} \right\}$	1)	6
b)	$V = \{A \mid A \in M_{3\times3}(\mathbb{R}),$ A is a symmetric matrix $\}$	ii)	$\left\{ \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 1 & 2 \\ 0 & 2 & 0 \\ 1 & 0 & 0 \end{bmatrix} \right\}$	2)	6
c)	$V = \{A \mid A \in M_{3\times3}(\mathbb{R}),$ A is an upper triangular matrix $\}$	iii)	$\left\{ \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ 1 & 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix} \right\}$	3)	2

Table : M2W4GA1

Choose the correct option.

- [ ] a → i → 2.
- [ ] b → i → 2.
- [x] a → ii → 3.
- [ ] c → iii → 1
- [ ] b → iii → 2.
- [ ] c → i → 1.

Partially Correct.

Score: 0.33

Accepted Answers:

a → ii → 3.
b → iii → 2.
c → i → 1.

3) Consider the vector space $V = \{\begin{bmatrix} a & b \\ b & c \end{bmatrix} \Big| \, a, b, c \in \mathbb{R}\}$. The subset $S = \{\begin{bmatrix} 2 & 1 \\ 1 & 3 \end{bmatrix}, \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}\}$ **1 point**

is a linearly independent subset of V . Which of the following matrices A are vectors in V such that $S \cup \{A\}$ is a basis for V ?

- [ ] $\begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix}$
- [x] $\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$
- [x] $\begin{bmatrix} 2 & 0 \\ 0 & 3 \end{bmatrix}$
- [ ] $\begin{bmatrix} 0 & 1 \\ 1 & 1 \end{bmatrix}$
- [x] $\begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix}$
- [ ] $\begin{bmatrix} 1 & -1 \\ -1 & 0 \end{bmatrix}$

Yes, the answer is correct.

Score: 1

Accepted Answers:

$\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$

$\begin{bmatrix} 2 & 0 \\ 0 & 3 \end{bmatrix}$

$\begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix}$

4) Let $S = \{v_1, v_2, v_3\}$ be a set of three vectors in $\mathbb{R}^4$. In order to determine the linear dependence/independence of S, we construct a matrix A whose i^{th} column is given by v_i, for $i \in \{1, 2, 3\}$. Let R denote the reduced row echelon form (RREF) of A, and let r_{ij} denote the $(i, j)^{th}$ entry of R. Choose the correct statements from the following. **1 point**

- [ ] R does not have any row comprising entirely of zeros if S is linearly independent.
- [x]

If k is the number of rows in R comprising entirely of zeros, then the dimension of Span(S) is given by 4 − k.

- [ ] If r_{11} and r_{23} are the pivots, then $\{v_1, v_2\}$ is a basis for Span(S).
- [x] If r_{11} and r_{23} are the pivots, then $\{v_1, v_3\}$ is a basis for Span(S).
- [x] rank(A) = rank(R), i.e., row operations do not alter the rank of a matrix.

Yes, the answer is correct.

Score: 1

Accepted Answers:

If k is the number of rows in R comprising entirely of zeros, then the dimension of Span(S) is given by 4 − k.
If r_{11} and r_{23} are the pivots, then $\{v_1, v_3\}$ is a basis for Span(S).
rank(A) = rank(R), i.e., row operations do not alter the rank of a matrix.

Numerical Answer Type (NAT):

5) Find the rank of the matrix A, where $A = [a_{ij}]$ is of order 2021×2021 and $a_{i,j} = \min\{i, j\}, i, j = 1, 2, \ldots 2021$
[**Hint:** First do it for smaller matrices, such as 3×3 or 4×4.]

2021

Yes, the answer is correct.

Score: 1

Accepted Answers:

(Type: Numeric) 2021

1 point

6) Consider a subspace $W = \{(x_1, x_2, x_3, x_4, x_5) \mid 8x_1 + 10x_5 = 0 \text{ and } 9x_2 + 6x_4 = 0\}$ of R^5. The dimension of W is

3

Yes, the answer is correct.
Score: 1

Accepted Answers:

(Type: Numeric) 3

1 point

7) Consider a matrix $A = \begin{bmatrix} 2 & -4 & 1 & 1 \\ 1 & -3 & 1 & -4 \\ 3 & -7 & 2 & 3 \end{bmatrix}$. The rank of the matrix is

3

Yes, the answer is correct.
Score: 1

Accepted Answers:

(Type: Numeric) 3

1 point

8) Consider two matrices $A = \begin{bmatrix} 3 & 9 & 8 \\ -7 & -10 & -9 \\ 2 & 6 & -7 \end{bmatrix}$ and $B = \begin{bmatrix} 3 & -7 & 2 \\ 9 & -10 & 6 \\ 8 & -9 & -7 \end{bmatrix}$. Then Rank$(A)$ - Rank (B) is

0

Yes, the answer is correct.
Score: 1

Accepted Answers:

(Type: Numeric) 0

1 point

9) Consider the subspace $V = \{(x, y, z) \mid z = 7x + 5y, \text{ where } x, y, z \in \mathbb{R}\}$ of $\mathbb{R}^3$ and span($\{(p, 0, 1), (0, q, 1)\}$) = V. Find the value of $\frac{1}{pq}$.

35

Yes, the answer is correct.

Score: 1

Accepted Answers:

(Type: Numeric) 35

1 point

10) Suppose V is a vector space defined as $V = \{A \mid A \in M_{4\times4}(\mathbb{R}),\ A$ is an upper triangular matrix, and the sum of the diagonal entries is zero $\}$. What is the cardinality of a basis of V?

9

Yes, the answer is correct.

Score: 1

Accepted Answers:

(Type: Numeric) 9.0

1 point

11) U is the vector space of all upper triangular matrices of order 3. V is the vector space of all lower triangular matrices of order 3. The dimensions of the vector spaces $\mathrm{U}, \mathrm{V}, \mathrm{U} \cap \mathrm{V}$ are a, b, c respectively. Find $a + b + c$. The usual rules of addition and scalar multiplication apply for all three spaces.

15

Yes, the answer is correct.

Score: 1

Accepted Answers:

(Type: Numeric) 15

1 point